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Studierichting: Informatica
Oefeningenreeks 1B nr. 39.7

Opgave: Bereken in $\mathbb{C}$ de vierdemachtswortel van -81.

Oplossing:

$-81 = (x+yi)^2$

Stel $(a+bi) = (x+yi)^2$. Dan $(a+bi)^2 = (x+yi)^4 = -81$

$-81 = (a+bi)^2 \Rightarrow a+bi = \pm 9i$

Als $\boldsymbol{a+bi=9i} = (x+yi)^2 = x^2 - y^2 + 2xyi$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 9 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -\dfrac{9^2}{4} \end{cases}$$

$$\lambda^2 - \frac{9^2}{4} = 0 \Rightarrow \lambda_{1,2} = \frac{\pm\sqrt{9^2}}{2} = \Big\langle \begin{array}{ll} +: & \dfrac{9}{2} = x^2 \Rightarrow x = \pm\dfrac{3}{\sqrt{2}} = \pm\dfrac{3\sqrt{2}}{2} \\ -: & -\dfrac{9}{2} = -y^2 \Rightarrow y = \pm\dfrac{3}{\sqrt{2}} = \pm\dfrac{3\sqrt{2}}{2} \end{array}$$

Dan is $x+yi \in \left\{\dfrac{3\sqrt{2}}{2} + \dfrac{3\sqrt{2}}{2}i, -\dfrac{3\sqrt{2}}{2} - \dfrac{3\sqrt{2}}{2}i\right\}$

Als $\boldsymbol{a+bi=-9i} = (x+yi)^2 = x^2 - y^2 + 2xyi$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = -9 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -\dfrac{9^2}{4} \end{cases}$$

$$\lambda^2 - \frac{9^2}{4} = 0 \Rightarrow \lambda_{1,2} = \frac{\pm\sqrt{9^2}}{2} = \Big\langle \begin{array}{ll} +: & \dfrac{9}{2} = x^2 \Rightarrow x = \pm\dfrac{3}{\sqrt{2}} = \pm\dfrac{3\sqrt{2}}{2} \\ -: & -\dfrac{9}{2} = -y^2 \Rightarrow y = \pm\dfrac{3}{\sqrt{2}} = \pm\dfrac{3\sqrt{2}}{2} \end{array}$$

Dan is $x+yi \in \left\{\dfrac{3\sqrt{2}}{2} - \dfrac{3\sqrt{2}}{2}i, -\dfrac{3\sqrt{2}}{2}i - \dfrac{3\sqrt{2}}{2}\right\}$

Oplossing: $x+yi \in \left\{\dfrac{3\sqrt{2}}{2} + \dfrac{3\sqrt{2}}{2}i, -\dfrac{3\sqrt{2}}{2} - \dfrac{3\sqrt{2}}{2}i, \dfrac{3\sqrt{2}}{2} - \dfrac{3\sqrt{2}}{2}i, -\dfrac{3\sqrt{2}}{2}i - \dfrac{3\sqrt{2}}{2}\right\}$

References